Cover Letter

Dear Editor,

We are pleased to submit our manuscript entitled **"Uncertainty propagation from incomplete tracking data in migratory-route comparison"** for consideration in *Movement Ecology*.

Animal tracking datasets often contain irregular sampling and prolonged observation gaps, yet migration routes are commonly compared after reducing each incomplete track to a single deterministic curve. This manuscript presents a validate–calibrate–propagate diagnostic framework for assessing how missing-route uncertainty propagates into trajectory-distance analyses, augmented with two new methodological contributions: state-conditioned perturbation scale estimation and split-conformal envelope calibration.

Using open GPS tracking data for Lesser Black-backed Gulls from the `LBBG_ZEEBRUGGE` dataset, we compare random point loss with contiguous tracking gaps, evaluate deterministic spherical bridge and Brownian bridge reconstruction, estimate behavioral-state-conditioned perturbation scales (resting, foraging, directed), apply split-conformal prediction to provide distribution-free finite-sample coverage guarantees, and propagate reconstruction uncertainty into distance matrices, clustering, anomaly ranking, and top-ranked anomaly overlap.

The key findings are:

1. Missingness mechanism matters more than missingness fraction: random thinning is largely absorbed by resampling, but contiguous gaps substantially alter downstream route-comparison conclusions.
2. Deterministic spherical bridge interpolation does not fully resolve the instability introduced by prolonged gaps.
3. A simple Brownian bridge baseline, with season-specific perturbation scales estimated from observed route residuals, makes reconstruction uncertainty visible but under-covers by a factor of 3–8×.
4. State-conditioned scale estimation reveals a twenty-fold range between resting and directed-flight perturbation magnitudes, but endpoint-state conditioning does not improve raw envelope calibration because gap-interior behavior is not predictable from boundary states alone.
5. Split-conformal calibration lifts empirical coverage from 33–66% to 80–94% with conformal quantiles of 3.8–4.0, providing a principled, distribution-free alternative to ad-hoc empirical inflation.

The paper is positioned as a movement ecology methods contribution. Its main claim is that route-comparison conclusions should not rely on a single deterministic distance matrix when prolonged tracking gaps are present. Instead, plausible reconstruction uncertainty should be validated and calibrated, then propagated into the downstream conclusions that movement ecologists inspect. To our knowledge, this is the first application of conformal prediction to animal trajectory uncertainty quantification.

We believe the manuscript fits *Movement Ecology* because it addresses a practical analytical challenge in movement data analysis—handling incomplete trajectories—and provides a transparent, reproducible framework for uncertainty-aware route comparison. The study engages with literature published in *Movement Ecology* and related journals on Brownian bridges, hidden Markov models, continuous-time movement models, and trajectory similarity analysis.

This manuscript is original, has not been published previously, and is not under consideration elsewhere.

The data used in the study are publicly available from the INBO `LBBG_ZEEBRUGGE` dataset. The analysis code and manuscript materials are available at:

https://github.com/cliffyb824/bird-migration-trajectory-metrics

The author declares no competing interests. This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Sincerely,

Yuan Qiu

Department of Statistics, College of Economics

Hangzhou Dianzi University

Hangzhou 310003, China

yuanqiu@hdu.edu.cn